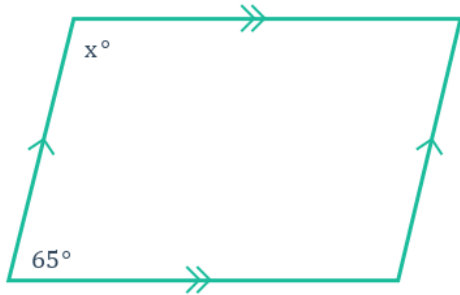


Properties of parallelograms



1 Find the value of x in each of the following parallelograms:

a



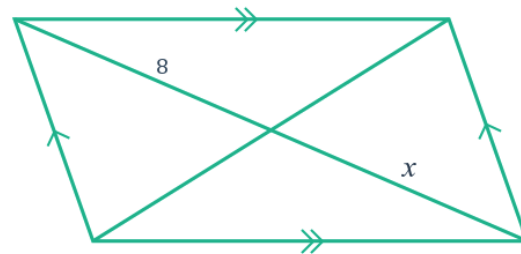
b



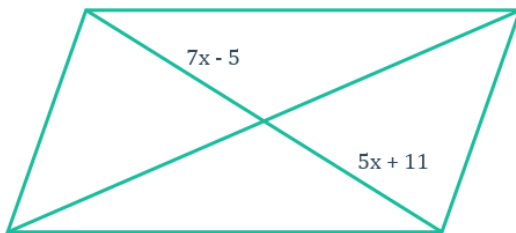
c



d



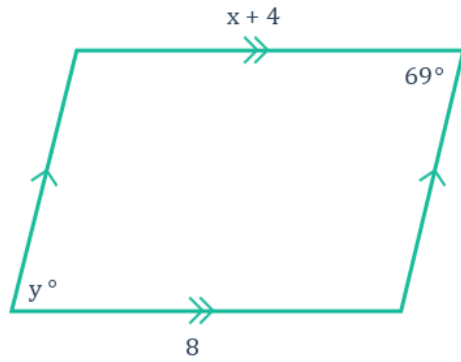
e



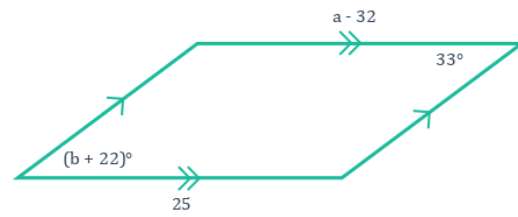
2 Find the value of the variables in the following diagrams:



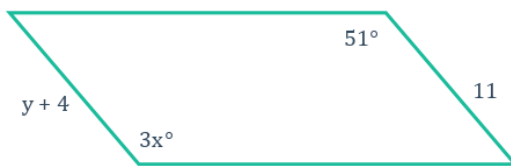
a



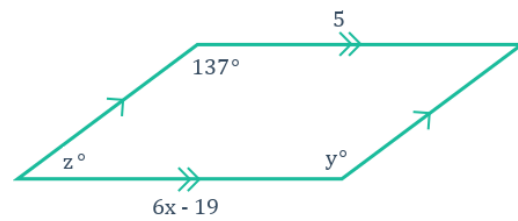
b



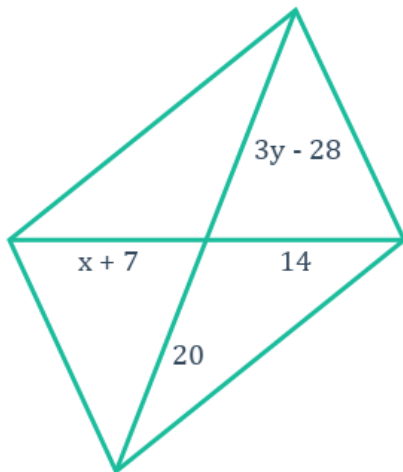
c



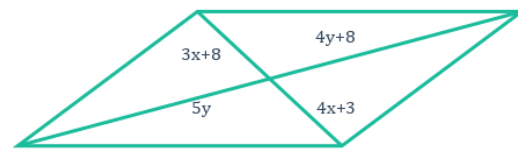
d



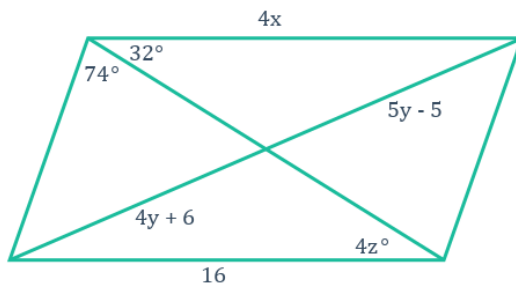
e



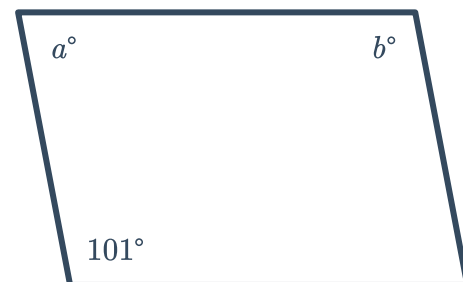
f



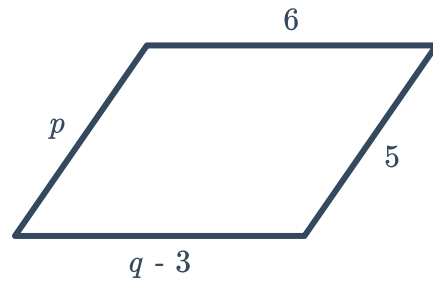
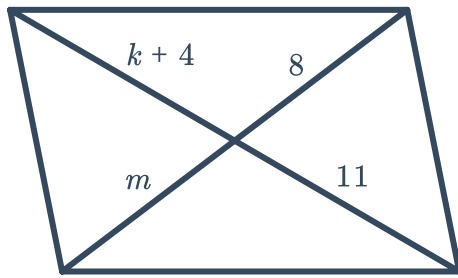
g



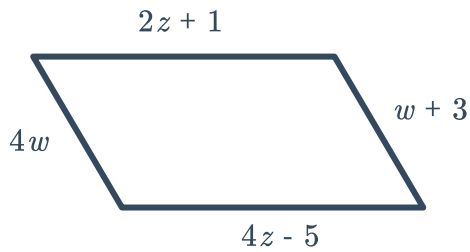
h



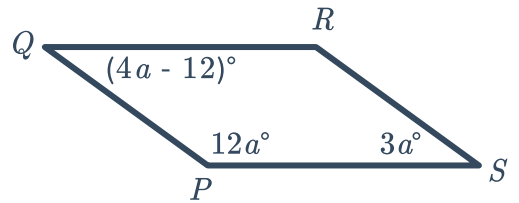
i



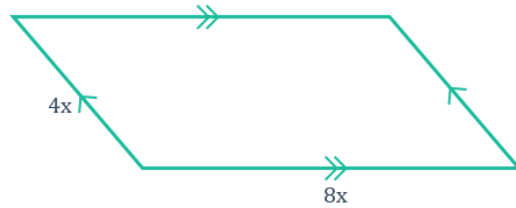
k



l

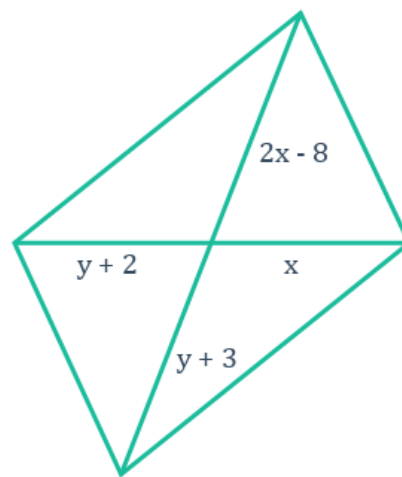


- 3 Find the value of x if the perimeter of the parallelogram given is 48 m.

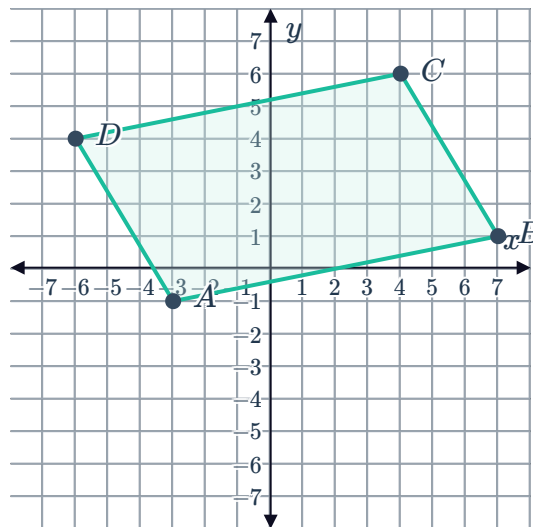


- 4 Consider the given parallelogram:

- a Find the values of x and y .
- b Determine the length of the longest diagonal.

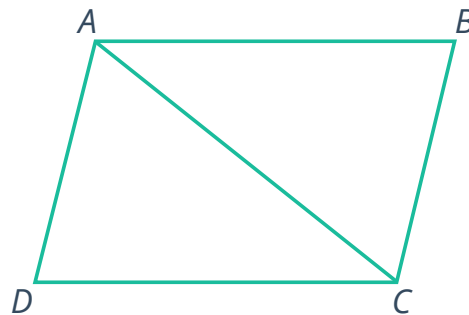


- 5 Consider the parallelogram $ABCD$ that has  graphed on the number plane.



- a Find the slope of sides:
- i $\overline{AB}$ and $\overline{CD}$
 - ii $\overline{AD}$ and $\overline{BC}$
- b One property of parallelograms is that opposite sides are parallel. State whether each of the following is a property of parallel lines.
- i They intersect.
 - ii Parallel lines have equal slopes.
 - iii Parallelograms are the only shapes in which parallel lines appear.
 - iv Two parallel lines have equal lengths.

- 6 Consider the following proof to show that the opposite sides of parallelogram $ABCD$ are congruent:



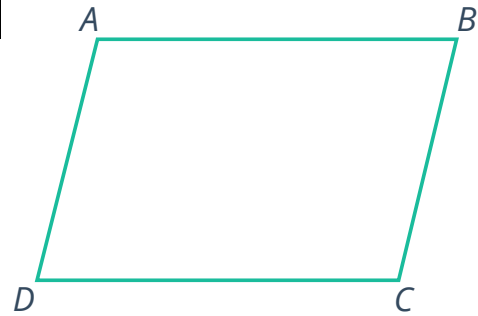
To prove: ?

Statements	Reasons
1. $ABCD$ is a parallelogram	Given
2. $\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
3. $\angle BAC \cong \angle DCA$ and $\angle BCA \cong \angle DAC$	Alternate Interior Angles Theorem
4. $\overline{AC} \cong \overline{AC}$	Reflexive property of congruence
5. $\triangle ABC \cong \triangle CDA$	Angle-Side-Angle Congruence
6. $\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$	Corresponding parts of congruent triangles are congruent (CPCTC)

Determine whether each of the following statements is shown to be true from the given proof.

- a If a quadrilateral is a parallelogram, then its opposite angles are congruent.
- b If a quadrilateral has opposite sides congruent, then it is a parallelogram.
- c If a quadrilateral is a parallelogram, then its opposite sides are congruent.
- d If a quadrilateral has consecutive interior angles supplementary, then its opposite sides are congruent.

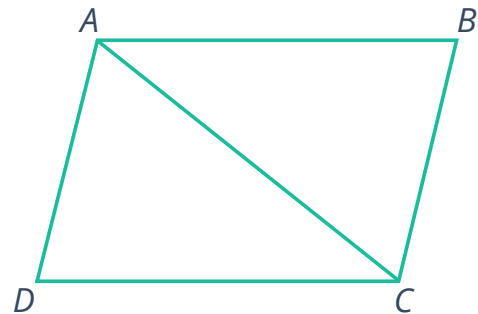
- 7 Given the parallelogram $ABCD$, write a statement that can be deduced from the proof.



To prove: ?

Statements	Reasons
1. $ABCD$ is a parallelogram	Given
2. $\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
3. $\angle A$ and $\angle B$, $\angle C$ and $\angle D$, and $\angle D$ and $\angle A$ are supplementary.	Consecutive interior angles postulate

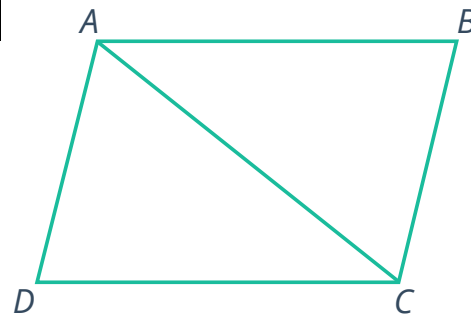
- 8 Given the parallelogram $ABCD$, complete the following two-column proof to show that $\triangle ABC$ is congruent to $\triangle CDA$:



To prove: $\triangle ABC \cong \triangle CDA$

Statements	Reasons
1. $ABCD$ is a parallelogram	Given
2. $\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
3. $\angle BAC \cong \angle DCA$ and $\angle BCA \cong \angle DAC$	Alternate Interior Angles Theorem
4. -	-
5. $\triangle ABC \cong \triangle CDA$	Angle-Side-Angle Congruence

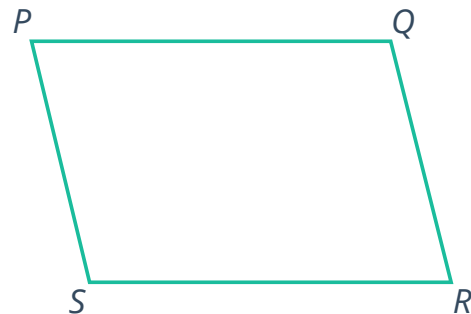
- 9 Given the parallelogram $ABCD$, complete the following two-column proof to show that the opposite sides are congruent:



To prove: Opposite sides of a parallelogram are congruent

Statements	Reasons
1. $ABCD$ is a parallelogram	Given
2. $\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
3. $\angle BAC \cong \angle DCA$ and $\angle BCA \cong \angle DAC$	Alternate Interior Angles Theorem
4. $\overline{AC} \cong \overline{AC}$	Reflexive property of congruence
5. -	-
6. $\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$	Corresponding parts of congruent triangles are congruent (CPCTC)

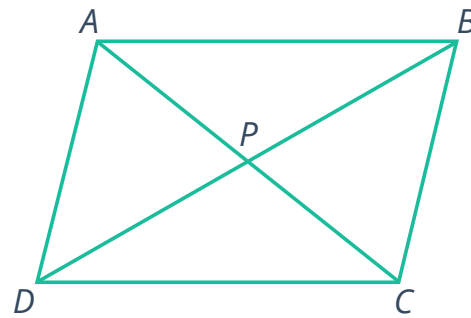
- 10 Given the parallelogram $PQRS$, complete the following two-column proof to prove that the opposite angles are congruent:



To prove: Opposite angles of a parallelogram are congruent.

Statements	Reasons
1. $PQRS$ is a parallelogram	Given
2. $\overline{PQ} \parallel \overline{SR}$ and $\overline{PS} \parallel \overline{QR}$	Definition of a parallelogram
3. $\angle P$ and $\angle S$, $\angle S$ and $\angle R$, and $\angle R$ and $\angle Q$ are supplementary.	Consecutive interior angles postulate
4. -	-

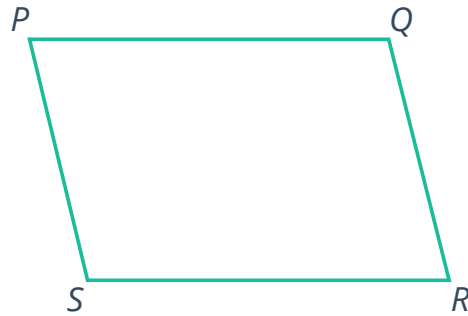
- 11 Given the parallelogram $ABCD$, complete the following two-column proof to show that the diagonals bisect each other:



To prove: Opposite sides of a parallelogram are congruent

Statements	Reasons
1. $ABCD$ is a parallelogram	Given
2. $\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
3. $\angle ABD \cong \angle CDB$ and $\angle ADB \cong \angle CBD$	Alternate Interior Angles Theorem
4. $\overline{BD} \cong \overline{BD}$	Reflexive property of congruence
5. $\triangle ABD \cong \triangle CDB$	Angle-Side-Angle congruence theorem
6. -	-
7. -	-
8. $\triangle ABP \cong \triangle CDP$	Angle-Side-Angle congruence theorem
9. $\overline{BP} \cong \overline{DP}$ and $\overline{AP} \cong \overline{CP}$	Corresponding parts of congruent triangles are congruent (CPCTC)
10. $\overline{AC}$ bisects $\overline{BD}$ and $\overline{BD}$ bisects $\overline{AC}$	Definition of a bisector

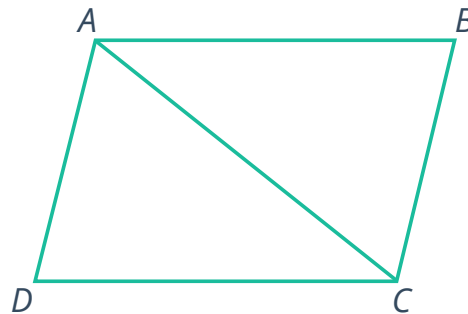
- 12 Given the parallelogram $PQRS$, complete the following two-column proof to prove that the consecutive angles are supplementary:



To prove: Consecutive angles of a parallelogram are supplementary

Statements	Reasons
1. $PQRS$ is a parallelogram	Given
2. -	-
3. -	-

- 13 Consider the two-colum proof to show that the opposite sides of the given parallelogram $ABCD$ are congruent:

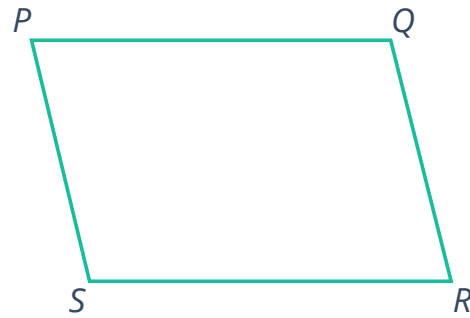


To prove: Opposite sides of a parallelogram are congruent

Statements	Reasons
1. $ABCD$ is a parallelogram	Given
2. $\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
3. $\angle BAC \cong \angle DCA$ and $\angle ACB \cong \angle CAD$	Alternate Interior Angles Theorem
4. $\overline{AC} \cong \overline{AC}$	Reflexive property of congruence
5. $\triangle ABC \cong \triangle CDA$	Angle-Side-Angle Congruence
6. $\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$	If a quadrilateral is a parallelogram, then it has both pairs of opposite sides congruent



- 14 Consider the two-column proof to show that the opposite angles of the given parallelogram $PQRS$ are congruent:



To prove: Opposite angles of a parallelogram are congruent.

Statements	Reasons
1. $PQRS$ is a parallelogram	Given
2. $\angle P$ and $\angle S$, $\angle S$ and $\angle R$, and $\angle R$ and $\angle Q$ are supplementary.	Consecutive interior angles postulate
3. $\overline{PQ} \parallel \overline{SR}$ and $\overline{PS} \parallel \overline{QR}$	Definition of a parallelogram
4. $\angle P \cong \angle R$ and $\angle Q \cong \angle S$	Congruent supplements theorem

State the error in the two-column proof.

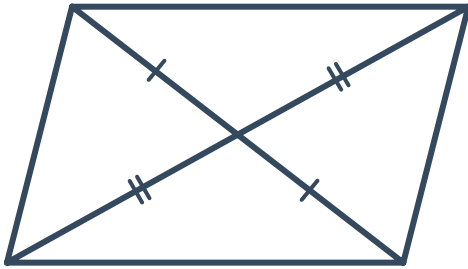
Additional questions

- 15 Determine whether each statement is true or false:
- a A parallelogram has two pairs of opposite sides that are congruent
 - b The diagonals of a parallelogram bisect the angles
 - c If the opposite angles of a quadrilateral are congruent, then the quadrilateral is a parallelogram
 - d The diagonals of a parallelogram bisect one another.
 - e A parallelogram has consecutive supplementary angles.
 - f A parallelogram has at least one right angle.

16 Based on the given markings, determine whether or not we can conclude the figure is a parallelogram:



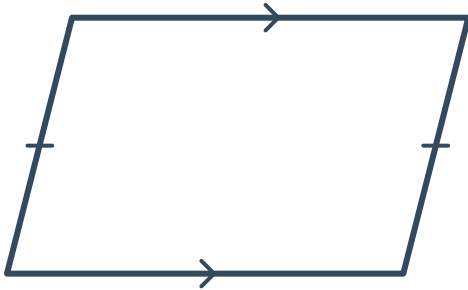
a



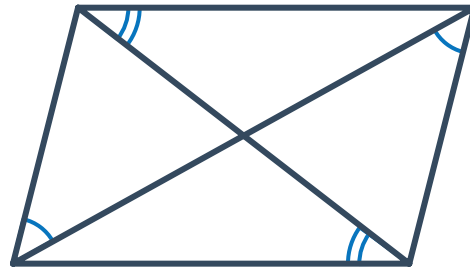
b



c

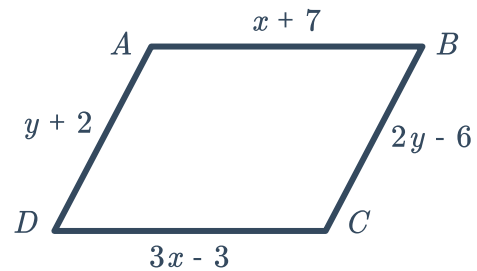


d



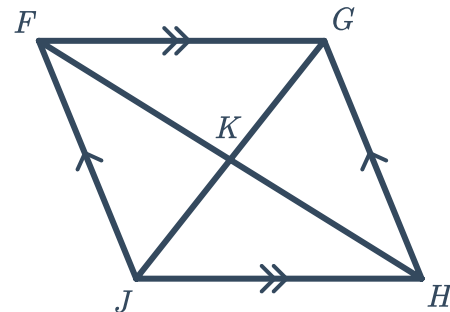
17 Given the parallelogram $ABCD$:

- a Solve for the variables.
- b Solve for the perimeter of $ABCD$.

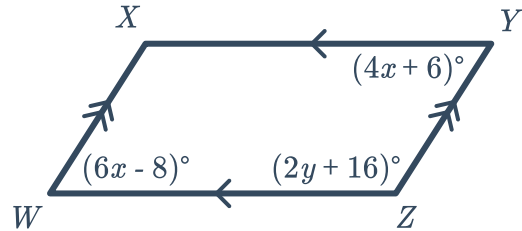


18 Given the parallelogram $FGHJ$:

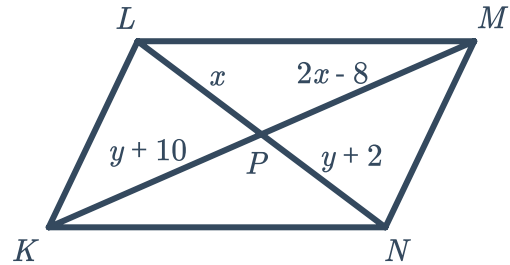
- a If $HF = 32$, find KF .
- b If $\angle F = 55^\circ$, find $\angle H$.
- c If $\angle G = 127^\circ$, find $\angle J$.



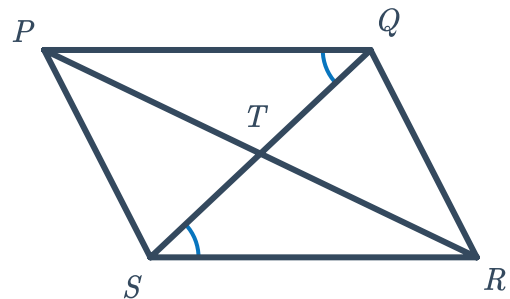
- 19 Find the value of y that makes $WXYZ$ a parallelogram.



- 20 Solve for LN in parallelogram $KLMN$.

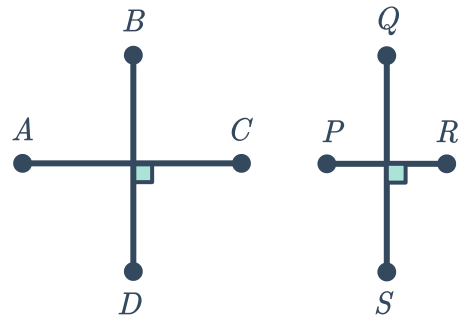


- 21
- What additional piece of information about the angles in $PQRS$ would allow you to prove it is a parallelogram?
 - What additional piece of information about the sides of $PQRS$ would allow you to prove it is a parallelogram?



- 22 Given: $\overline{AC} \perp \overline{BD}$.
 $\overline{PR} \perp \overline{QS}$

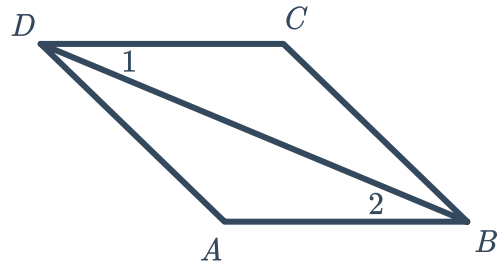
- Copy the diagrams and connect the endpoints in alphabetical order, forming two quadrilaterals.
- Is $ABCD \cong PQRS$? Explain how you know.
- What information would be required for conclude that $ABCD$ and $PQRS$ to be classified as parallelograms?



23 Given: $\overline{AB} \cong \overline{CD}$

$\angle 1 \cong \angle 2$

Prove: $ABCD$ is a parallelogram



24 Rani is constructing a garden in his yard. He plans to build a fence around the garden in the shape of a quadrilateral. Rani has three design options in mind for his garden:

- Option 1: The diagonals of the garden are perpendicular.
- Option 2: One pair of opposite corners form 35° angles, while the other pair of opposite corners form 125° angles.
- Option 3: One pair of opposite sides are 8 ft. long.

Will any of Rani's garden designs form a parallelogram? Explain how you know.

25 Given: Parallelogram $ABCD$

Prove: E is the midpoint of $\overline{AC}$

